

# Jiayi Zhang

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## Education

### University of Nottingham Ningbo China (UNNC)

BSc (HONS) MATHEMATICS WITH APPLIED MATHEMATICS

- GPA - 3.83/4.00

Ningbo China

Sep 2024 - Jun 2028 (Expected)

## Research Interests

My research accelerates **decision-making** and **path planning** for embodied agents that **perceive**, **reason**, and **interact** in the physical world. I study how **multimodal world models** couple with **search**, **planning**, and **policy learning—reinforcement learning** (model-based, offline), **imitation learning**, and **learning-to-search**—to make reliable choices under uncertainty and limited compute. Grounded in **mathematics**, **AI**, and **robotics**, I aim for real-time **embodied intelligence** that plans, adapts, and acts safely in complex environments.

## Experience

### Tsinghua University (THU)

RESEARCH ASSIST AT [TEA LAB](#), ADVISED BY [PROF. HUAZHE XU](#)

- **Research Topics:** Robotics, Embodied AI

Beijing, China

October 2025

### Shanghai Jiao Tong University (SJTU)

RESEARCH ASSIST AT [IWIN-FINS LAB](#), ADVISED BY [PROF. JIANPING HE](#)

- **Research Topics:** optimal control, generative model, grasping

Shanghai, China

June 2025 - Octo 2025

### The University of Hong Kong (HKU)

RESEARCH INTERN AT [IoT LAB](#), ADVISED BY [PROF. EDITH C. H. NGAI](#)

- **Research Topics:** federated learning, fairness, spatio-temporal graphs.

Hong Kong SAR, China

March 2025 – August 2025

## Publications

### PREPRINTS

#### [P1] F<sup>2</sup>STNet: Fair and Federated Spectral-Temporal Modeling for Graph Forecasting

Jiayi Zhang\*, Jinfeng Xu\*, Hewei Wang\*, Siyuan Cen, Haidong Huang, Yiyao Zhan, Zheyu Chen, Jinjiang You, Ai Jian, Edith C. H. Ngai  
Under review, 2025

### CONFERENCE PROCEEDINGS

#### [C4] Multi-modal Dynamic Proxy Learning for Personalized Multiple Clustering

Jinfeng Xu, Zheyu Chen, Shuo Yang, Jinze Li, Ziyue Peng, Zewei Liu, Hewei Wang, [Jiayi Zhang](#), Edith C. H. Ngai  
AAAI, 2026

#### [C3] Multi-Cali Anything: Dense Feature Multi-Frame Structure-from-Motion for Large-Scale Camera Array Calibration

Jinjiang You\*, Hewei Wang\*, Yijie Li, Mingxiao Huo, LVT Ha, Mingyuan Ma, Jinfeng Xu, [Jiayi Zhang](#), Puzhen Wu, Shubham Garg, Wei Pu  
IROS, 2025 (Oral) [\[Paper\]](#)

#### [C2] Spec-LLAVA: Accelerating Vision-Language Models with Dynamic Tree-Based Speculative Decoding

Mingxiao Huo\*, [Jiayi Zhang](#)\*, Hewei Wang\*, Jinfeng Xu, Zheyu Chen, Huilin Tai, Ian Yijun Chen  
ICML Workshop, 2025 [\[Paper\]](#)

#### [C1] Segregation and Context Aggregation Network for Real-Time Cloud Segmentation

Yijie Li\*, Hewei Wang\*, [Jiayi Zhang](#), Jinjiang You, Jinfeng Xu, Puzhen Wu, Yunzhong Xiao, Soumyabrata Dev  
ICLR Workshop, 2025 [\[Paper\]](#)

### JOURNAL ARTICLES

#### [J2] TrafficKAN-GCN: Graph Convolutional-based Kolmogorov-Arnold Network for Traffic Flow Optimization

[Jiayi Zhang](#)\*, Yiming Zhang\*, Yuan Zheng, Yuchen Wang, Jinjiang You, Yuchen Xu, Wenxing Jiang, Soumyabrata Dev  
arXiv preprint, 2025 [\[Paper\]](#)

#### [J1] RAINER: A Robust Ensemble Learning Grid Search-Tuned Framework for Rainfall Patterns Prediction

Zhenqi Li\*, Junhao Zhong\*, Hewei Wang, Jinfeng Xu, Yijie Li, Jinjiang You, [Jiayi Zhang](#), Runzhi Wu, Soumyabrata Dev  
arXiv preprint, 2025 [\[Paper\]](#)

## Honors & Awards

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2025    **Honorable Mention**, Interdisciplinary Contest in Modeling 2025

Online

## Academic Services

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**Reviewer**    NeurIPS, AAAI, AISTATS

2025–Present

## Leadership/Mentorship/Social

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### Semester Peer Mentor

UNIVERSITY OF NOTTINGHAM NINGBO CHINA (UNNC)

- Led class building and fostered a positive community culture.
- Provided academic guidance and practical resources for freshmen.
- Supported peer integration and promoted social connectedness.

Ningbo China

Sep. 2025–Jun. 2026

## Skills

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**Programming**    Python, C++,  $\text{\LaTeX}$ , MATLAB, HTML

**Frameworks**    PyTorch, Tensorflow, NumPy, Flask, MySQL, Git, Anaconda, OpenCV, ROS

**Models**    CNN, ResNet, Faster R-CNN, YOLO, LSTM, Transformer, BERT, ViT, GANs, VAE, Stable Diffusion

**Platforms**    Unitree Z1 Pro, Franka Research 3

## Additional

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**Language**    Chinese (native), English (working proficiency)

**Interests**    Running, hiking, mahjong, chessing, and comics